

Uncertainty-Aware Scan-to-BIM via Evidential Deep Learning

A Calibrated Decision Layer for Autonomous IFC Updates in Robot-Mounted Digital-Twin Maintenance

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Abstract

Robot-mounted Scan-to-BIM systems for autonomous digital-twin maintenance must decide, in real time, when a detected discrepancy between sensor evidence and the as-designed Building Information Model warrants an autonomous IFC update, when it should be deferred to a human operator, and when it should be ignored as sensor noise. Recent agentic-AI frameworks such as BIM2RDT [1] demonstrate the perception and event-injection portions of this loop but explicitly leave the geometric-update closure as future work. We propose to close this gap by inserting a calibrated uncertainty layer between point-cloud semantic segmentation and IFC editing. A segmentation network with an evidential deep learning head produces per-point Dirichlet-distributed predictive uncertainty in a single forward pass; a decision policy then maps the joint signal of per-element uncertainty and scan-vs-BIM geometric discrepancy onto an action set comprising autonomous update, human-in-the-loop review, missing-element annotation, and ignore, with autonomous BIM mutation gated on the conjunction of low uncertainty and a discrepancy exceeding the scanner noise floor. To the best of our knowledge, this is the first application of evidential deep learning to indoor and AEC-domain point-cloud semantic segmentation, and the first coupling of per-point evidential uncertainty to a decision policy for autonomous IFC element updating. Our proposal includes a two-phase proof of concept to validate the segmentation and uncertainty-calibration components on the S3DIS benchmark and demonstrate the decision policy on a synthetic Scan-vs-BIM scenario constructed with HELIOS++ from a small IFC model. The proposed perception system is embedded in a quadruped-patrol deployment scenario that addresses the four fundamental robotics pillars — mapping, localization, planning, and control — through established off-the-shelf components, leaving the calibrated decision layer as the sole novel contribution.

1. Project Overview

Project Title: Uncertainty-Aware Scan-to-BIM via Evidential Deep Learning: A Calibrated Decision Layer for Autonomous IFC Updates.

Application Area: Reality-to-BIM scan matching, robot-mounted construction monitoring, and autonomous digital-twin maintenance.

Core Problem: A robot scanning a construction site can detect that a physical element disagrees with the as-designed BIM, but cannot decide what

to do about the disagreement in any principled manner. Existing systems lack a calibrated trust signal mediating between perception and BIM mutation, with the result that automated IFC updates are committed only in narrow, domain-specific settings using fixed geometric thresholds and no quantified perception confidence.

Why Automation Matters:

- Manual Scan-to-BIM workflows for medium commercial buildings take two to four weeks and cost in the low tens of thousands of dollars per build-

ing [2].

- Continuous as-built updates are presently unaffordable, leaving deployed BIMs systematically out-of-date relative to the assets they represent.
- Robot-mounted perception is the natural data-acquisition layer for continuous monitoring, but the autonomy boundary — when does the robot commit, when does it ask for help, when does it ignore? — has no principled formalization in current systems.
- A calibrated uncertainty layer enables conservative autonomy: the system commits only when both perception and geometry agree, and defers to a human operator otherwise, making the autonomy boundary auditable.

2. Introduction

Building Information Models (BIMs) have become the standard digital representation of architectural assets across their design, construction, and operation phases, yet keeping a BIM synchronized with the physical asset it represents remains an unresolved engineering problem. Manual Scan-to-BIM workflows for medium commercial buildings are estimated by industry vendors to take two to four weeks and cost in the low tens of thousands of dollars [2], while recent deep-learning-based Scan-to-BIM pipelines have pushed automated 3D-element-level precision into the 96–99% range [3]. The research frontier has accordingly shifted from *initial* Scan-to-BIM conversion — which is now a largely solved engineering problem at the pipeline-quality threshold — to *continuous* digital-twin maintenance: the question of how a BIM model can be kept synchronized with a physical asset as that asset evolves through construction, renovation, or operation. Robot-mounted point-cloud capture, in which a mobile robot autonomously scans a site and updates the BIM in response to the scans, is the most demanding instance of this problem and the one this proposal targets.

The most recent and most directly relevant prior work in this thread is BIM2RDT [1], an agentic-AI framework for robot-ready site digital twins. BIM2RDT introduces Semantic-Gravity ICP — a registration algorithm that uses LLM-derived gravity

priors to constrain the alignment of robot-acquired scans against the reference BIM and demonstrates an end-to-end pipeline in which IoT-sensor events are written into the IFC representation of the BIM via `IfcTask` and `IfcEvent` entities. What the framework does not specify or evaluate is the *geometric* update closure: the conversion of a detected discrepancy between scan and BIM into a geometric edit of the IFC element. The authors explicitly identify “error-correction mechanisms for imperfect BIM inputs” and “real-time data synchronization and conflict resolution” as future work. This omission is field-recognized rather than specific to BIM2RDT — recent reviews of digital-twin maintenance in the AEC industry consistently identify real-time BIM-reality synchronization as an unresolved open problem [4, 5].

The methodological obstacle to closing this loop is not segmentation accuracy. Patil and Kalantari [6], in a comparative study of five recent point-cloud segmentation backbones on a Scan-to-BIM workflow, find that all models exceeding approximately 65% mean intersection-over-union produce reliable downstream BIM reconstructions — a threshold cleared by every modern transformer-based architecture. The methodologically interesting research question is therefore no longer how to achieve higher segmentation accuracy, but how to know when a segmentation can be trusted enough to act on autonomously. Existing AEC point-cloud segmentation pipelines produce per-point class predictions as point estimates without quantified confidence; an autonomous BIM-update system that consumes these predictions has no principled basis for choosing between committing an update, deferring to a human operator, and ignoring an apparent discrepancy as sensor noise. This is the gap the present proposal addresses.

We propose **the first uncertainty-aware Scan-to-BIM framework, applying single-forward-pass evidential deep learning to indoor point-cloud semantic segmentation and coupling per-point uncertainty to a decision policy for autonomous IFC element updating**. The framework consists of (i) a point-cloud segmentation network with a Dirichlet-parameterized evidence head [7] producing per-point predictive uncertainty in a single forward pass, (ii) a scan-vs-BIM geometric

matching layer that computes per-point distances to a reference IFC model, and (iii) a decision policy that maps the joint signal of per-element uncertainty and geometric discrepancy onto an action set comprising autonomous update, human-in-the-loop review, missing-element annotation, and ignore.

The perception system is embedded in a quadruped-patrol deployment scenario in which the four fundamental robotics pillars are addressed through established off-the-shelf components: *mapping* via a tightly-coupled lidar-inertial SLAM stack (LIO-SAM [8]) that produces both a 2D occupancy grid for navigation and per-patrol 3D point clouds for perception; *localization* via the same LIO-SAM stack with Scan Context [9] loop closure and global re-localization, anchored to the BIM frame by a one-time SG-ICP alignment at the docking station; *planning* via the ROS 2 Navigation Stack (Nav2) [10] with A^* [11] on the occupancy grid and behavior-tree orchestration of waypoint patrols; and *control* via velocity commands to the quadruped platform’s on-board locomotion controller. Treating these pillars as engineering scaffolding rather than as research contributions allows the proposal to focus on the calibrated decision layer between perception and IFC editing, which is where the methodological novelty lies.

3. Literature Review and Motivation

The proposal sits at the intersection of three established research threads. We summarize each below and identify the gap each thread leaves open.

3.1 Prior Work Summary

- **Point-cloud semantic segmentation in AEC.** Modern transformer-based backbones — Point Transformer V3 [12], Swin3D [13], PointMetaBase [14] — now exceed 70% mIoU on the S3DIS Area 5 benchmark [15], and AEC-specific pipelines such as Mahmoud *et al.* [3] and DawNet [16] achieve 96–99% precision at the 3D-element level. Key finding [6]: above ~65% mIoU, segmentation accuracy is no longer the bottleneck for downstream BIM reconstruction.
- **Scan-vs-BIM discrepancy detection and**

IFC updating. The geometric-matching tradition reaches its mature form in Hu and Brilakis [17], achieving mIoU 0.914 across five real-world datasets via a PROSAC-based pipeline. A small but real literature implements automated IFC mutation in narrow domains: pose updates for steel framing [18], parameter updates for pipeline systems [19], pose and deformation updates for structural columns [20], and progress-status property-set writes [21]. *None of these systems gate the IFC mutation on calibrated perception uncertainty.*

- **Uncertainty quantification for 3D segmentation.** Sampling-based methods — MC Dropout [22], deep ensembles [23] — multiply inference cost by the sample count. Evidential deep learning (EDL) [7] produces per-point uncertainty in a single forward pass via a Dirichlet output head, and has been applied to outdoor LiDAR panoptic segmentation [24, 25]. The closest AEC-UQ precedent, Vassilev *et al.* [26], uses Bayesian KPConv on bridges — not buildings — and does not couple uncertainty to a downstream decision policy.

3.2 Differentiating Features of Proposed System

The proposal is positioned along three axes against the closest prior art:

1. **Method:** single-forward-pass evidential uncertainty rather than the multi-pass Bayesian sampling of [26]. This is dictated by the real-time robot-mounted constraint — T -fold inference overhead is incompatible with on-robot deployment.
2. **Domain:** indoor building scenes with IFC-mappable class structure rather than the outdoor LiDAR [24, 25] or bridge-infrastructure [26] regimes where evidential and Bayesian UQ have previously appeared in 3D.
3. **Downstream coupling:** an explicit IFC-edit decision policy that maps (uncertainty, discrepancy) to an action set, rather than treating segmentation uncertainty as a terminal output.

The contribution is the *combination*. We do not

claim novelty for EDL itself, for EDL applied to point clouds (precedent in [24, 25]), for AEC-domain UQ (precedent in [26]), or for the abstract pattern of uncertainty-gated robotic decisions (precedent in [27, 28]). We claim novelty for the specific combination, restricted to the specific domain of indoor BIM-coupled robot-mounted Scan-to-BIM. See Table 1 for a comparison of this work with the closest prior art.

4. Problem Statement and Automation Opportunity

Problem Statement: Given a robot-acquired point cloud $\mathbf{X} = \{p_1, \dots, p_N\}$ of an indoor scene and a reference Building Information Model $\mathcal{B} = \{e_1, \dots, e_M\}$ as a set of IFC elements, determine for each element e_j an action $a_j \in \{\text{ignore}, \text{update}, \text{flag-for-review}, \text{mark-missing}\}$ such that autonomous BIM mutations are committed only when perception confidence and geometric evidence jointly support the action, and the decision is otherwise deferred to a human operator.

Automation Opportunity: Three steps in the existing manual Scan-to-BIM-update workflow are good candidates for intelligent automation: (i) per-point semantic classification of the scan, currently performed visually by a trained technician at hours per room; (ii) per-element discrepancy detection between scan and reference BIM, currently performed by manual overlay in CAD software; and (iii) the decision of which discrepancies warrant a BIM edit versus operator review. The first two are increasingly addressed by the automation pipelines surveyed in [3, 16], but the third remains unaddressed. The proposal automates the third step by introducing a calibrated decision policy whose threshold parameters are operator-tunable.

4.1 Inputs and Outputs

Inputs:

- Point cloud $\mathbf{X}$ from a robot-mounted TLS, MLS, or RGB-D sensor (XYZ, optionally RGB and surface normals).
- Reference IFC-format BIM $\mathcal{B}$ with element-level class labels.

- Scanner specification (nominal range error σ_{range} , point density ρ).
- Operator-specified policy thresholds ($c_{\min}, d_{\text{translate}}, u_{\text{lo}}, u_{\text{hi}}$) and utility weights.

Outputs:

- Per-element action $a_j \in \mathcal{A}$ with traceable per-element signals $(c_j, \bar{u}_j, \bar{d}_j)$.
- Updated BIM $\mathcal{B}'$ with property-set annotations on every element and geometric edits on **update**-action elements.
- Audit log of decisions and supporting signals for human review.

5. System-Level Overview

The proposed system is a five-layer perception-to-action pipeline embedded within a quadruped-patrol robotics operational sequence, as depicted in Figure 1. The robotics scaffolding (top row of Figure 1) handles the four fundamental robotics pillars — mapping, localization, planning, and control — and delivers a registered point cloud to the perception/decision pipeline at each scan waypoint; the technical depth of these components is given in §5.2. The perception/decision pipeline (bottom row of Figure 1) is the methodological core of this proposal. Layer 1 (Evidential Semantic Segmentation) consumes the point cloud and produces per-point class predictions and per-point predictive uncertainties via a single forward pass through a network with a Dirichlet output head. Layer 2 (Scan-vs-BIM Geometric Matching) associates each scan point with its nearest IFC element and computes a per-point distance to the reference BIM surface. Layer 3 (Per-Element Aggregation) reduces per-point signals to per-element scalars: coverage, mean discrepancy, and mean uncertainty. Layer 4 (Decision Policy) maps the per-element triple to a discrete action through a deterministic partition of signal space. Layer 5 (IFC Mutation) realizes each action as an IfcOpenShell API operation, with autonomous geometric mutation localized to the **update** action only.

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Table 1: Comparison with closest prior art.

Work	UQ Method	Domain	IFC Update?
BIM2RDT [1]	None	Indoor + outdoor	Events only
Vassilev [26]	Bayesian	Bridges	No
EvLPSNet [24]	Evidential	LiDAR / outdoor	No
ULOPS [25]	Evidential	LiDAR / outdoor	No
This work	Evidential	Indoor / IFC	Yes (policy)

5.1 Operational Workflow

The complete patrol cycle, mapping each step to a stage in Figure 1, is as follows:

1. **Mission start (Docking Station):** the robot departs a fiducial-marked docking station whose pose in the BIM frame was registered via a one-time SG-ICP alignment.
2. **Navigation (Move to Waypoint):** a Nav2-based global planner computes a path to the next scan waypoint over the 2D occupancy grid, and the platform’s onboard locomotion controller executes the path under DWB-based local obstacle avoidance.
3. **Mapping and localization (Localize & Map):** LIO-SAM’s lidar-inertial Extended Kalman Filters (EKF) provides drift-bounded continuous localization en route, with Scan Context loop closure correcting accumulated drift on the closed patrol trajectory.
4. **Sensing (LiDAR Scan):** on arrival at the waypoint, the robot performs a stationary 3D LiDAR scan, producing a point cloud $\mathbf{X}$ registered to the BIM coordinate frame.
5. **Perception:** Layer 1 performs evidential semantic segmentation on the point cloud and produces predictive probability and uncertainty $(\hat{\mathbf{p}}_i, u_i)$ per point in a single forward pass.
6. **Discrepancy estimation:** Layers 2–3 produce per-element coverage, mean discrepancy, and mean uncertainty $(c_j, \bar{u}_j, \bar{d}_j)$ from the point cloud and the reference BIM $\mathcal{B}$.
7. **Decision:** Layer 4 produces per-element action $a_j \in \mathcal{A}$, representing a decision to **update**, **flag-for-review**, **mark-missing**, or **ignore** IFC element j .

8. **Execution and logging:** Layer 5 applies the action via IfcOpenShell, producing updated BIM $\mathcal{B}'$ and audit log.

9. **Patrol-loop continuation:** the robot returns to a waypoint (or to the docking station for charging) and the cycle repeats, accumulating a temporal record of as-built deviations and update events across many scans.

5.2 Deployment Scenario and Robotics Integration

The perception system is deployed on a quadruped robot (Boston Dynamics Spot or equivalent: Unitree Go2, ANYmal) conducting periodic patrol-based digital-twin maintenance of an operational facility — for example, an office building undergoing internal reconfiguration or a research facility where partition walls and fixtures change between projects. Each patrol traverses a closed loop of scan waypoints, performs a stationary LiDAR scan at each waypoint, and returns to a docking station. The patrol-loop framing aligns with the proposal’s continuous digital-twin maintenance thesis and produces a temporal record of as-built deviations and update events across many scans. The sensor stack is 3D LiDAR (Velodyne VLP-16 or Livox Mid-360) as the primary perception sensor, an RGB-D camera (Intel RealSense D435i) for close-range obstacle avoidance, and a 9-DoF IMU for high-rate state estimation.

Mapping. The system maintains two complementary map artifacts. A 2D occupancy grid is built incrementally during patrol from LiDAR returns and serves as the cost map for navigation. A 3D point cloud is captured per-patrol and registered against the as-designed BIM frame — not against a previous scan, since cumulative SLAM would propagate drift into the BIM frame. LIO-SAM [8] provides the lidar-

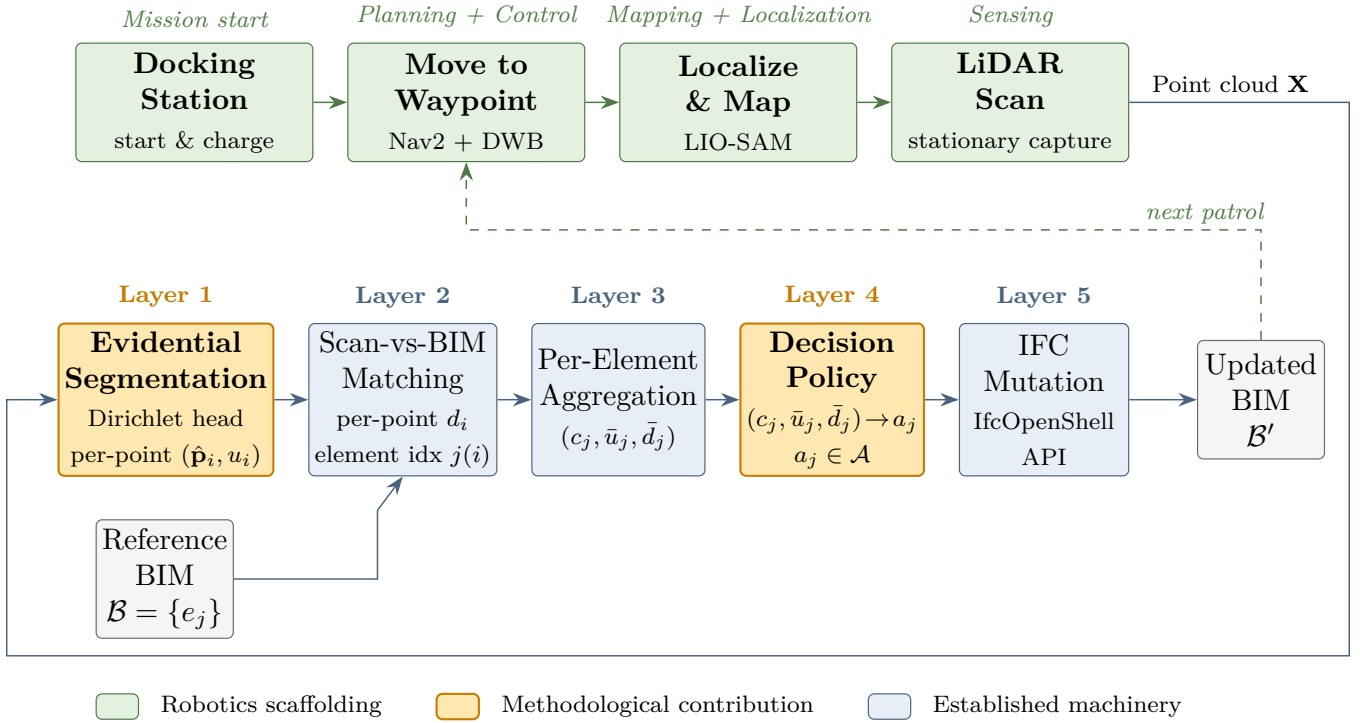


Figure 1: Operational pipeline of the proposed framework. The top row shows the robotics operational sequence executed during a patrol: the quadruped departs the docking station, navigates to a scan waypoint via Nav2-based planning and the platform’s onboard locomotion controller, performs LIO-SAM-based localization and mapping en route, and captures a stationary LiDAR scan at the waypoint. The resulting point cloud $\mathbf{X}$ is handed off to the bottom row, which executes the proposal’s perception and decision pipeline: the point cloud and reference BIM $\mathcal{B}$ are processed through five sequential layers, producing per-element actions $a_j \in \mathcal{A}$ that are realized as IfcOpenShell operations on the updated BIM $\mathcal{B}'$. Layers 1 and 4 (highlighted) are the load-bearing methodological contributions of this proposal; Layers 2, 3, and 5 use established machinery. The dashed arrow indicates the patrol-loop continuation: after the BIM is updated, the robot returns to the next waypoint and the cycle repeats.

inertial SLAM front-end for both artifacts. Initial alignment to the BIM frame is performed once at the docking station via SG-ICP [1] against fiducial-marked geometry, and all subsequent patrols inherit this alignment.

Localization. LIO-SAM’s tightly-coupled lidar-IMU EKF provides drift-bounded continuous localization; an extended Kalman filter fuses 200 Hz IMU predictions with 10 Hz LiDAR scan-to-scan corrections via the standard preintegration formulation [29]. Loop closure on the closed patrol trajectory is provided by Scan Context [9], which also handles the kidnapped-robot problem: unexpected pose discontinuities trigger a global re-localization against the prior 3D map from a single LiDAR scan. SG-ICP docking re-alignment provides the last-resort fallback if Scan Context fails after substantial site

changes. Sensor noise is handled within the LIO-SAM pipeline and is treated as an upstream concern by the perception layer; localization residuals are bounded by the LIO-SAM drift envelope.

Planning. The system uses the ROS 2 Navigation Stack (Nav2) [10] for both global and local planning. Scan waypoints are specified offline based on the BIM’s room geometry and sequenced into a closed loop via a TSP nearest-neighbor heuristic with 2-opt refinement. Paths between successive waypoints are computed by A^* [11] on the 2D occupancy grid, which is preferred over sampling-based planners (RRT, RRT*) for this fully-observed low-dimensional setting. The DWB local planner handles dynamic obstacles encountered during patrol execution. The full patrol behavior is orchestrated by a Nav2 behavior tree with standard recovery behaviors

(clear costmap, spin in place, back up).

Control. The local planner emits 2D body-frame velocity commands (v_x, v_y, ω_z) at 10 Hz to the quadruped platform’s onboard locomotion controller, which translates them into per-joint torques, gait selection, foot placement, and balance behavior. No custom control development is performed; the platform vendor’s controller is treated as a black-box capability with a clean velocity-command interface. This delegation is justifiable because low-level quadruped locomotion is itself a deep research area whose engineering would dwarf the proposal’s perception contribution without adding methodological value.

The integration interface between the robotics stack and the perception layer is narrow: the robot delivers BIM-frame point clouds with localization covariance, and the perception layer’s decision policy conservatively downgrades **update** to **flag-for-review** when localization uncertainty exceeds the discrepancy threshold $d_{\text{translate}}$. Table 2 catalogs the robotics-pillar tooling.

6. Mathematical Formulation

6.1 Variables and Notation

- $\mathbf{X} = \{p_1, \dots, p_N\}$: input point cloud, $p_i \in \mathbb{R}^d$, $d \in \{3, 6, 9\}$.
- $\mathcal{B} = \{e_1, \dots, e_M\}$: reference BIM as a set of IFC elements; $S(e_j)$ is the surface of element e_j .
- K : number of segmentation classes; $c(e_j)$: class associated with element e_j ’s IFC class.
- f_{θ} : segmentation network with parameters θ , producing per-point per-class evidence $e_{i,k} \geq 0$ for point p_i and class $k \in \{1, \dots, K\}$.
- α (Dirichlet concentration parameters): in evidential deep learning, the per-class evidence values are converted to the concentration parameters of a Dirichlet distribution over the K -class probability simplex. Specifically:
 - $\alpha_{i,k} = e_{i,k} + 1$: per-point per-class concentration. Adding 1 to the evidence reflects a uniform Dirichlet prior; a point with no evidence ($e_{i,k} = 0$ for all k) yields $\alpha_i = \mathbf{1}$, recovering the maximum-uncertainty uniform prior.

- $\alpha_i = (\alpha_{i,1}, \dots, \alpha_{i,K})$: per-point concentration *vector*; the K -dimensional parameter that fully specifies the Dirichlet distribution $\text{Dir}(\alpha_i)$ over class probabilities for point p_i .
- $\alpha_{i,0} = \sum_{k=1}^K \alpha_{i,k}$: Dirichlet *strength*, the sum of per-class concentrations. Interpretable as one plus the total accumulated evidence; large $\alpha_{i,0}$ means the network has strong evidence for some class assignment, while $\alpha_{i,0} = K$ means no evidence has been accumulated.
- $\tilde{\alpha}_i$: per-point *KL-target* concentration vector, defined as α_i with the target-class concentration reset to 1, i.e. $\tilde{\alpha}_{i,k} = 1$ if k is the ground-truth class for p_i and $\tilde{\alpha}_{i,k} = \alpha_{i,k}$ otherwise. Used in the KL-divergence regularizer of Equation (2) to penalize evidence on *non-target* classes without penalizing evidence on the correct class.
- $\hat{p}_{i,k} = \alpha_{i,k}/\alpha_{i,0}$, $u_i = K/\alpha_{i,0}$: per-point predictive probability for class k (the Dirichlet mean) and per-point predictive uncertainty (the *vacuity*, i.e. residual belief mass uncommitted by the available evidence).
- $d_i, j(i)$: per-point distance to nearest BIM surface and index of nearest element.
- $c_j, \bar{d}_j, \bar{u}_j$: per-element coverage, mean discrepancy, and mean uncertainty.
- $a_j \in \mathcal{A}$: per-element action.

6.2 Objective Function

The segmentation network is trained to minimize the KL-regularized sum-of-squares Bayes-risk loss [7]:

$$\theta^* = \arg \min_{\theta} \sum_n \sum_i \mathcal{L}_{\text{EDL}}(\alpha_i^{(n)}, \mathbf{y}_i^{(n)}) \quad (1)$$

$$\mathcal{L}_{\text{EDL}} = \sum_{k=1}^K \left[(y_k - \hat{p}_k)^2 + \frac{\hat{p}_k(1 - \hat{p}_k)}{\alpha_0 + 1} \right] + \lambda_t \cdot \text{KL}(\text{Dir}(\tilde{\alpha}) \parallel \text{Dir}(\mathbf{1})) \quad (2)$$

where $\tilde{\alpha}$ is α with the target-class concentration reset to 1 and $\lambda_t = \min(1, t/T_{\text{anneal}})$ with $T_{\text{anneal}} = 10$ epochs. The annealing is mandatory; without it, the network converges to a degenerate uniform-evidence solution.

Table 2: Robotics stack mapping (deployment).

Pillar / Layer	Tool / Algorithm
Platform	Boston Dynamics Spot / Unitree Go2 / ANYmal
LiDAR	Velodyne VLP-16 / Livox Mid-360
RGB-D	Intel RealSense D435i
IMU	9-DoF (RealSense-integrated or BNO055)
Mapping (3D + 2D)	LIO-SAM [8]
BIM-frame alignment	SG-ICP [1] (one-time)
Localization	LIO-SAM EKF + IMU preintegration [29]
Loop closure / re-localization	Scan Context [9]
Global planner	A* [11] on 2D occupancy grid
Local planner	DWB (Nav2 default)
Orchestration	Nav2 behavior trees [10]
Control	Platform vendor SDK (velocity-command interface)
Middleware	ROS 2 Humble

6.3 Constraints

Per-point geometric matching:

$$d_i = \min_{s \in S(\mathcal{B})} \|p_i - s\|_2, \quad d_i \leq d_{\max} \quad (3)$$

Per-element aggregation:

$$c_j = \frac{|P_j|}{N_j^{\text{exp}}}, \quad \bar{d}_j = \frac{\sum_{p_i \in P_j} d_i}{|P_j|}, \quad \bar{u}_j = \frac{\sum_{p_i \in P_j} u_i}{|P_j|} \quad (4)$$

Decision-policy thresholds satisfy $u_{\text{lo}} < u_{\text{hi}}$ and $d_{\text{translate}} > 3\sigma_{\text{range}}$.

System-level constraints: total inference time per scan $\leq T_{\text{frame}}$ (real-time deployment); decision rule is deterministic for auditability; default action under undefined inputs is **flag-for-review** (conservative-action principle).

6.4 Uncertainty Sources

The system operates under six distinct uncertainty sources:

- **(U1) Aleatoric segmentation uncertainty:** captured by the predictive entropy $H(\hat{\mathbf{p}}_i)$ of the Dirichlet mean.
- **(U2) Epistemic segmentation uncertainty:** captured by the vacuity $u_i = K/\alpha_{i,0}$, the primary signal used by the decision policy.
- **(U3) Scanner range and registration error:** characterized by σ_{range} ; addressed by setting

$$d_{\text{translate}} > 3\sigma_{\text{range}}.$$

- **(U4) Coverage estimation uncertainty:** N_j^{exp} is itself estimated via synthetic-scan calibration; conservative default biases toward **mark-missing**.
- **(U5) Threshold calibration uncertainty:** addressed by the sensitivity analysis of §11.
- **(U6) BIM ground-truth uncertainty:** the reference BIM is treated as oracle; this is the most fundamental open methodological problem and is conservatively addressed by the policy’s **flag-for-review** default but not formally modeled.

7. Methodology

7.1 Perception / Sensing Module

The segmentation backbone is **PointMetaBase-L** [14], chosen for its 2.7M-parameter footprint (compatible with the 8GB VRAM ceiling of the deployment GPU) and its 69.7% S3DIS Area 5 mIoU — which clears the Patil and Kalantari [6] reconstruction-sufficient threshold by approximately five points. The standard softmax classification head is replaced by a softplus-activated evidence head producing $e_{i,k} \geq 0$, with Dirichlet concentration $\alpha_{i,k} = e_{i,k} + 1$. The vacuity $u_i = K/\alpha_{i,0}$ is the per-point trust signal consumed by the decision policy. Single-pass inference is the architectural property that makes real-time deployment feasible: sampling-based alternatives (MC Dropout at $T = 20$, deep

ensembles at $N = 3$) would multiply segmentation inference cost by $20\times$ and $3\times$ respectively.

7.2 Decision / Planning / Optimization Module

The decision policy maps $(c_j, \bar{u}_j, \bar{d}_j)$ to $a_j \in \mathcal{A}$ via a coverage pre-check followed by a partition of the $(\bar{u}_j, \bar{d}_j)$ plane. The Algorithm 1 pseudocode specifies the rule. The policy is deliberately asymmetric: **update** — the only action that mutates IFC geometry — requires the conjunction of low uncertainty and a discrepancy exceeding the geometric threshold. Every ambiguous configuration falls through to **flag-for-review** or **ignore**. This conservatism is the operational expression of the trust-calibration argument: in any condition where either signal is ambiguous, the system defers rather than acts.

Algorithm 1 Decision policy.

```

1: Input:  $(c_j, \bar{u}_j, \bar{d}_j)$  for each element  $e_j$ ; thresholds  $(c_{\min}, d_{\text{translate}}, u_{\text{lo}}, u_{\text{hi}})$ 
2: Output: action  $a_j \in \mathcal{A}$ 
3: for each IFC element  $e_j$  do
4:   if  $c_j < c_{\min}$  then
5:      $a_j \leftarrow \text{mark-missing}$ ; continue
6:   end if
7:   if  $\bar{u}_j \geq u_{\text{hi}}$  then
8:      $a_j \leftarrow \text{flag-for-review}$ 
9:   else if  $\bar{u}_j < u_{\text{lo}}$  and  $\bar{d}_j \geq d_{\text{translate}}$  then
10:     $a_j \leftarrow \text{update}$ 
11:   else if  $\bar{u}_j < u_{\text{lo}}$  and  $\bar{d}_j < d_{\text{translate}}$  then
12:     $a_j \leftarrow \text{ignore}$ 
13:   else
14:     if  $\bar{d}_j \geq d_{\text{translate}}$  then
15:        $a_j \leftarrow \text{flag-for-review}$ 
16:     else
17:        $a_j \leftarrow \text{ignore}$ 
18:     end if
19:   end if
20: end for

```

The policy instantiates a research pattern — *uncertainty-gated rejection of the autonomous default action* — established in adjacent robotics literatures, including evidential rejection-option logic [25], conformal-prediction safety thresholds [27], and conditional autonomy in surgical robotics [28].

Threshold calibration: $c_{\min}$ is set as the 5th percentile of the validation coverage distribution; $d_{\text{translate}}$ is set at $3\sigma_{\text{range}}$ ($\sim 3\text{--}5$ cm for a Riegl VZ-400 TLS); u_{lo} and u_{hi} are calibrated against held-out validation data using either percentile-based defaults or accuracy-anchored thresholds (largest u at which validation accuracy exceeds 95%; smallest at which it drops below 70%).

7.3 Learning or Inference Module

The segmentation network is trained on the standard S3DIS protocol [15]: train on Areas 1, 2, 3, 4, 6; test on Area 5. Three random seeds are run for the softmax variant of PointMetaBase-L (these constitute the deep-ensemble baseline); one seed for the EDL variant; one seed for temperature-scaled softmax (with T fit on a held-out validation slice by NLL minimization [30]).

Training stability requires three implementation commitments [7, 31]: softplus activation on the evidence head (not ReLU — avoids dead-evidence units); KL annealing per Equation (2); numerical stability via `logsumexp` computation of $\log \alpha_0$ and evidence clipping at 10^3 during early training.

Inference: single forward pass produces $(\hat{\mathbf{p}}_i, u_i)$ for each point. Per-point distances d_i are computed via Open3D’s KD-tree against IfcOpenShell-sampled BIM surfaces. Per-element aggregation and policy evaluation are deterministic.

8. Hardware Stack

8.1 Required Hardware

- **Sensors:** Riegl VZ-400 or comparable terrestrial LiDAR (TLS) for benchmark-quality acquisition; Velodyne VLP-16 or comparable mobile LiDAR (MLS) for robot-mounted deployment; iPad Pro LiDAR or comparable RGB-D for low-cost inspection passes.
- **Platform:** for the proof of concept, a single-workstation simulation environment; for eventual deployment, a quadruped robot (e.g., Boston Dynamics Spot, Unitree Go) carrying the perception and compute stack.
- **Compute:** development and training on an

NVIDIA RTX 3070 (8 GB VRAM, eGPU via OCuLink); 64 GB system RAM; Ubuntu 24.04. Production inference on equivalent or stronger embedded GPU (e.g., NVIDIA Jetson Orin) co-located with the robot platform.

- **Communication:** wireless LAN for telemetry and remote operator review; ROS 2 middleware for sensor and actuator integration during deployment.
- **Power / Support:** robot-onboard battery and protective sensor housing during deployment; standard workstation power for development.

9. Software Stack

9.1 Core Software Components

- **Programming Languages:** Python 3.10 (primary); C++ where IfcOpenShell or HELIOS++ require it.
- **Frameworks / Middleware:** PyTorch 2.3 with CUDA 12.1, PyTorch Geometric 2.5, Open3D 0.18, IfcOpenShell 0.8.
- **Simulation Tools:** HELIOS++ [32] for synthetic LiDAR data generation from IFC models; SPC scaffolding [33] for IFC-to-cloud pipeline reuse.
- **Optimization / Estimation Libraries:** SciPy for temperature-scaling fit; `torchmetrics` or `netcal` for ECE computation.
- **Integration / Deployment:** Git for version control, Weights & Biases for experiment tracking, `conda environment.yml` for reproducibility.

Table 3: Hardware/software stack mapping.

Layer	Stack
Sensing (deployment)	TLS / MLS / RGB-D
Sensing (PoC)	HELIOS++ synthetic [32]
Segmentation	PyTorch + PyG, Point-MetaBase
EDL head	Custom, after [7]
Geometry	Open3D + IfcOpenShell
Decision policy	Pure Python
IFC mutation	IfcOpenShell API
Review interface	Bonsai (BlenderBIM)

10. Uncertainty Quantification

Uncertainty handling is the central methodological commitment of this proposal. The system replaces standard softmax classification — which produces miscalibrated overconfidence in modern deep networks [30] — with an evidential head producing per-point predictive uncertainty in a single forward pass.

10.1 Sources of Uncertainty

The six uncertainty sources U1–U6 enumerated in §6.4 cover the full uncertainty surface of the system: aleatoric and epistemic perception uncertainty, scanner noise, coverage estimation error, threshold calibration error, and BIM ground-truth uncertainty.

10.2 Handling Uncertainty

- **Evidential deep learning** [7]: per-point Dirichlet distribution over class probabilities; $u_i = K/\alpha_{i,0}$ has a principled interpretation under subjective logic [34] as the residual belief mass uncommitted by the available evidence.
- **Calibration measurement** [30]: Expected Calibration Error and reliability diagrams; temperature scaling as the canonical cheap calibration baseline.
- **Conservative decision policy** (§7.2): the autonomy boundary is gated on the conjunction of low uncertainty and confirmed geometric discrepancy; ambiguous configurations route to `flag-for-review`.

- **Operator-tunable thresholds:** the four policy thresholds reflect deployment-specific risk tolerance and are calibrated against held-out validation data.

10.3 Evaluation Metrics

- **Segmentation accuracy:** mean Intersection-over-Union (mIoU) on S3DIS Area 5.
- **Calibration:** Expected Calibration Error (ECE), Maximum Calibration Error (MCE), Negative Log-Likelihood (NLL), reliability diagrams.
- **Misclassification AUROC:** per-point uncertainty as the score, per-point misclassification as the binary label — the metric most directly relevant to the BIM-update decision policy.
- **Wall-clock inference time:** per-method, on a fixed test scan, averaged over 10 runs after warmup. EDL’s headline advantage over deep ensembles is single-pass inference; this metric quantifies it.

11. Proof of Concept and Application

11.1 Prototype or Test Scenario

The proof of concept is structured in two stages, with the second stage conditional on the establishment of baseline and core models.

Baseline stage (establish baseline and core models): environment bootstrap; PointNet++ [35] baseline reproduction (target $\geq 55\%$ Area 5 mIoU); PointMetaBase-L training (target $\geq 67\%$); evidential head replacement and training (target within 1–2 mIoU of softmax baseline); calibration evaluation against four UQ baselines (vanilla softmax, temperature-scaled softmax, 3-seed deep ensemble, MC Dropout where applicable). The baseline stage produces a calibration table comparing five UQ methods on five metrics plus inference time.

Sythentic Scann-vs-BIM stage (full pipeline verification): synthetic Scan-vs-BIM scenario via HELIOS++ [32] from a small known-good IFC model with one element deliberately displaced by ≥ 30 cm; trained EDL model run over the synthetic scan; per-element signals computed; decision policy applied. Success: policy produces **update** or

flag-for-review on the displaced element and **ignore** on a control element. Technical challenges encountered with the HELIOS++ pipeline development triggers fallback to the SPC dataset [33] with manually corrupted element.

A sensitivity analysis sweeps each of the four decision-policy thresholds across a meaningful range, defending the policy against the criticism that the thresholds are arbitrary.

11.2 Application Scenario

The intended deployment scenario is robot-mounted Scan-to-BIM on active construction sites or operating facilities, in which a quadruped robot carrying a LiDAR or RGB-D sensor periodically scans the asset, and the proposed framework processes each scan to identify and act on per-element discrepancies. The autonomy boundary is auditable and operator-tunable: the system commits autonomously only when both perception and geometry agree; otherwise it queues the discrepancy for human review. Operational value lies in reducing the cost of continuous as-built BIM maintenance from weeks-per-building manual conversion to per-scan automated reconciliation.

11.3 Expected Results

- Calibration table showing EDL non-dominated against the four UQ baselines at single-pass inference cost.
- Per-point uncertainty heatmap showing high uncertainty concentrated at class boundaries and in occluded regions.
- Decision policy correctly firing on the synthetic discrepancy scenario, with traceable per-element signal logging.
- Sensitivity sweep showing action-distribution stability under threshold perturbation.

12. Discussion

The proposal’s empirical claims are deliberately scoped. The proof of concept operates on synthetic point clouds and on the S3DIS benchmark rather

than on real construction-site scans; any extension to live deployment requires additional validation. The IFC mutation operations are sketched in IfcOpen-Shell code but not invoked live during the proof of concept; a closed-loop demonstration in which the BIM is mutated and a subsequent scan processed against the updated BIM is the natural next experimental step. Training and evaluation are restricted to a single benchmark, and cross-building generalization is not addressed. The reference BIM is treated as the geometric ground truth in the discrepancy computation — an idealization that the decision policy mitigates conservatively through its **flag-for-review** default but does not formally address.

Practical deployment considerations include the regulatory question of autonomous BIM mutation in safety-critical contexts (the **update** action is a write operation on the as-built record); the audit-trail requirement that every action — including **ignore** — produces a property-set entry timestamping the decision and supporting signals; and the integration question of how the **flag-for-review** branch surfaces to a human operator (Bonsai/BlenderBIM is the natural review interface).

13. Conclusion

This proposal has presented an uncertainty-aware Scan-to-BIM framework that closes the perception-to-action loop left open by recent robot-mounted digital-twin maintenance systems. The framework replaces the standard softmax classification head of a point-cloud segmentation network with a Dirichlet-parameterized evidence head producing per-point predictive uncertainty in a single forward pass, computes per-element scan-vs-BIM geometric discrepancies, and applies a deterministic decision policy that maps the joint signal of per-element uncertainty and discrepancy onto an action set comprising autonomous update, human-in-the-loop review, missing-element annotation, and ignore. The contribution is positioned along three axes against the closest prior art: single-pass evidential rather than multi-pass Bayesian uncertainty, indoor-building rather than outdoor-LiDAR or bridge-infrastructure domain, and an explicit IFC-edit decision policy rather than segmentation as a terminal output.

The most immediate future-work extension is closure of the loop into live IFC mutation. Cross-dataset evaluation on ScanNet or BIMNet tests cross-building generalization without retraining the substantive contribution. Methodological refinements include Posterior Networks [36] addressing the asymptotic-uncertainty critique [37] and conformal-prediction calibration of decision-policy thresholds [27] replacing percentile-based threshold setting with distribution-free correctness guarantees. The most fundamental open problem — and the one this proposal explicitly does not address — is uncertainty about the reference BIM itself; a system that maintains a probability distribution over the BIM rather than treating it as oracle is a substantial research undertaking but is the right destination for autonomous digital-twin maintenance research.

The four fundamental robotics pillars are addressed by the deployment scenario through established off-the-shelf components: *mapping* maintains a 2D occupancy grid and per-patrol BIM-frame point clouds via LIO-SAM [8]; *localization* couples LIO-SAM’s lidar-inertial EKF with Scan Context [9] loop closure and re-localization, with SG-ICP [1] as the docking-station fallback; *planning* uses the ROS 2 Navigation Stack [10] with A^* [11] on the occupancy grid and behavior-tree orchestration of waypoint patrols; and *control* delegates to the quadruped platform’s on-board locomotion controller via a velocity-command interface. The robotics layer is treated as engineering scaffolding rather than as a research contribution, allowing the proposal’s methodological emphasis to remain on the calibrated decision layer between perception and IFC editing.

Robot-mounted Scan-to-BIM systems have reached the point where perception, registration, and event injection are well-understood components, and the binding constraint on autonomous digital-twin maintenance has shifted from accuracy to trust. The framework advanced here addresses that constraint by inserting a calibrated uncertainty layer between segmentation and IFC editing, positioned not as a terminal contribution but as the first concrete experiment in a research program whose direction the broader field has begun to recognize but has not yet pursued.

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